ML OPTIMIZATION

Quantization & Pruning Visual Guide

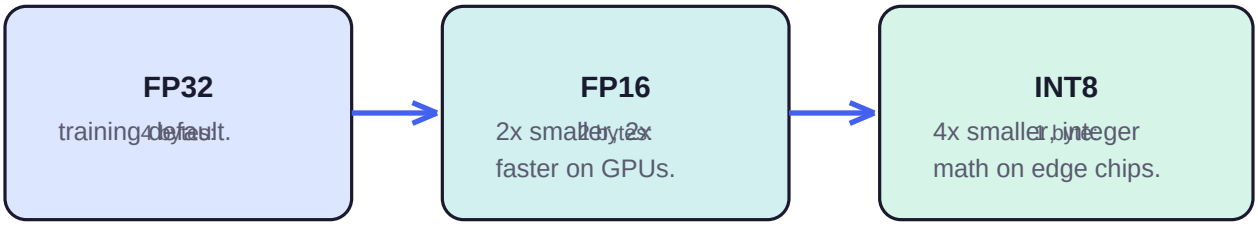
Smaller, faster models:
INT8/FP16, PTQ vs QAT, pruning, distillation, edge.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Precision Ladder

Fewer bits per weight: smaller, faster, nearly as good.



Memory bandwidth — not FLOPs — usually limits inference. Halving precision roughly doubles speed.

STEP 2/6

PTQ vs QAT

Quantize after training, or train quantization-aware.

PTQ

post-training: quantize a finished model with calibration data. Fast.

QAT

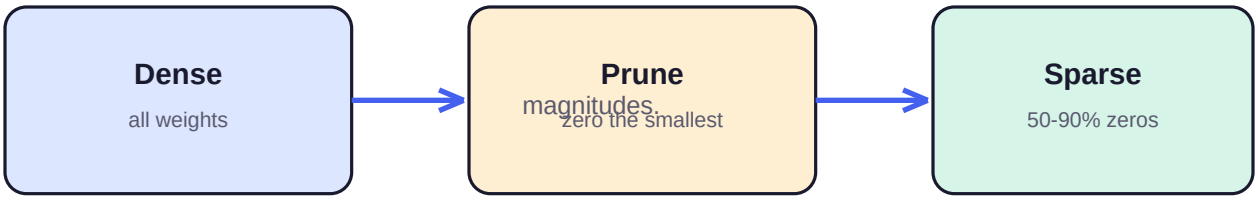
quantization-aware: simulate quantization during training. Slower, better.

Start with PTQ — it recovers most of the accuracy for free. Reach for QAT when PTQ degrades quality.

STEP 3/6

Magnitude Pruning

Delete the weights that matter least.



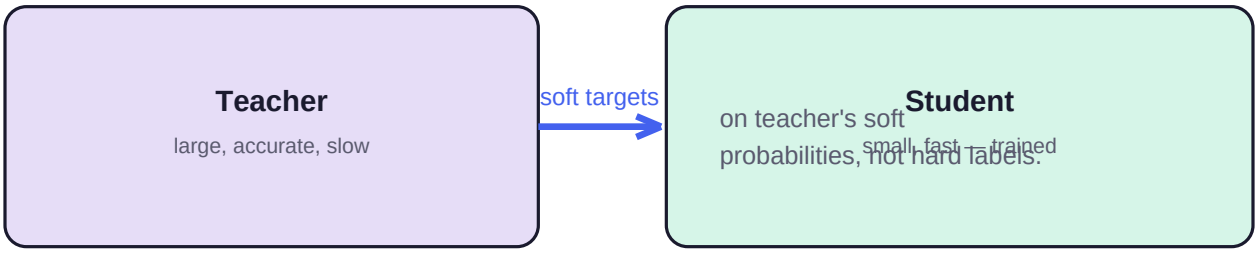
Iterative prune-and-fine-tune beats one-shot pruning. Unstructured sparsity needs special kernels to pay off.

Structured pruning (whole channels/filters) gives real speedups on standard hardware; unstructured needs sparse libraries.

STEP 4/6

Knowledge Distillation

A small student learns from a big teacher.

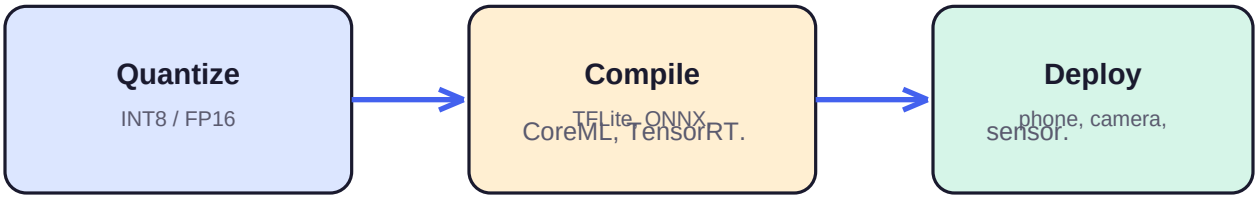


Soft targets carry 'dark knowledge' — which wrong answers were almost right. Students often beat same-size models trained from scratch.

STEP 5/6

Edge Deployment Stack

From cloud model to on-device.



Measure on the actual device — desktop benchmarks lie about mobile NPUs and DSPs.

STEP 6/6

The Accuracy Tradeoff

How much quality does compression cost?

- FP16: essentially free — do it always for inference.
- INT8 PTQ: usually <1% accuracy loss on vision and many NLP tasks.
- Aggressive pruning/distillation: expect 1-3% loss; validate on YOUR data, not the paper's.
- Combine techniques: distill, then quantize — the gains stack.
- Always keep the full-precision model as the reference for regression tests.

STEP ✓/✓

Key Takeaways

Quantization & Pruning — Remember

- FP16 is free; INT8 gives 4x compression with tiny accuracy cost.
- PTQ first (fast); QAT when PTQ isn't enough.
- Prune small magnitudes iteratively; prefer structured pruning for real speedups.
- Distillation transfers dark knowledge from teacher soft targets to a small student.
- Measure on the target device; keep full-precision as the regression reference.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev